

Japan Tech Tour Takeaways

Michelle Jing

SFC CE No. BUK594

michellejing@gfgroup.com.hk

Jeff Pu, CFA

SFC CE No. BNO719

jeffpu@gfgroup.com.hk

Henry Huang

SFC CE No. BUK594

henryhuang@gfgroup.com.hk

What's new

During our recent tech tour in Japan, we visited Mitsubishi Electric, Fuji Electric, TEL, Advantest, Nikon, and TOK. Below we summarize our key observations and insights.

Comments

Japanese companies adopt a conservative pricing stance, with TEL as the notable exception: Most companies either refrained from price hikes or limited to cost pass-through, except **TEL** which has implemented surcharges on current models and is pushing price increases on new models, targeting 50% GPM within 1–2 years (vs. FY2026 guidance of 45.5%). **TOK** is also facing solvent cost inflation and implemented pass-throughs in May at a notably faster pace than its historical norm. Fuji Electric, Mitsubishi Electric, Advantest, and Nikon all maintain relatively conservative pricing strategy, opting for cost pass-through on select products, though it's difficult to fully pass through.

800VDC development progressing well; SiC adoption accelerating: **Mitsubishi Electric** has commenced shipments of CDUs for 750kW to 1.2MW, and its joint development with NVIDIA on 800VDC power and cooling systems is advancing smoothly. Its newly built 8-inch SiC fab is now in production, with explosive data center SiC demand expected. **Fuji Electric** forecasts auto demand recovery by 2028, with SiC penetration reaching ~60% of auto power semi by 2030 (vs. ~25% currently).

SPE demand robust: **TEL** saw strong Korea and Taiwan region demand for etching tools (HARC, interconnect, GAA), driving etching production from one to two shifts. Advanced packaging tool demand (advanced logic/HBM) is expected to grow 60% YoY in FY2027. **Advantest** plans to expand SoC tester capacity to 10k units annually by 2028 and is confident in maintaining its 65% market share. For **China WFE**, TEL forecasts 20% YoY growth in 2026 driven by memory, logic foundries, and emerging customers; despite a CY2025 revenue decline due to share loss against US peers, TEL is confident in regaining share in 2026.

Geopolitical risks have proven less severe than initially feared: **TEL** and **Advantest** expect a stable China revenue mix (low-30% for TEL, ~20% for Advantest). **TOK** saw Chinese customers build inventory to 4–5 months, causing a temporary 2Q dip but no lasting impact expected. **Nikon** noted ArF dry products are unrestricted; advanced immersion tools can be sold with downgraded specs for China, with China contribution expected in the low-20% range.

Risks

1) AI demand slowing down; 2) Geo-political risks; 3) Competition.

Mitsubishi Electric

Datacenter cooling: Mitsubishi Electric has commenced shipments of its newly developed Cooling Distribution Unit (CDU), which leverages liquid cooling technology. In parallel, the company is collaborating with Zutacore on a heat exchanger-based system designed to cool gaseous media generated from heated liquids, with commercial rollout targeted for fiscal 2028. For fiscal 2025, the company's IT cooling business generated ¥32 billion in revenue. With the addition of its liquid cooling portfolio, Mitsubishi Electric projects revenue to exceed ¥100 billion by fiscal 2031, and further reach ¥300 billion by fiscal 2036.

Figure 1: Mitsubishi Electric's datacenter related products

Power supply systems

- Providing comprehensive systems, including UPSs^{*2}, switchboards, and batteries
- Revenue: FY26: 86.5 billion yen ⇒ **FY31: 180.0 billion yen+**



UPS

IT cooling systems

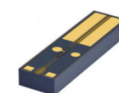
- Providing integrated total cooling solutions for data centers
- Revenue: FY26: 32.0 billion yen ⇒ **FY31: 100.0 billion yen+**



Cooling systems

Optical devices

- Providing ultra-high-speed optical devices (EML chips^{*3}) that support the expansion of the generative AI market
- Revenue: FY26: 40.0 billion yen ⇒ **FY31: 80.0 billion yen**



EML chips

Monitoring and control systems and factory automation-related components

- Providing industrial mechatronics products for applications, such as monitoring and control systems and component processing for cooling equipment
- Revenue: FY26: 20.0 billion yen ⇒ **FY31: 40.0 billion yen**



Programmable logic controllers (PLCs)

CNC^{*4}

Sources: Company Data, GF Securities (Hong Kong) Brokerage.

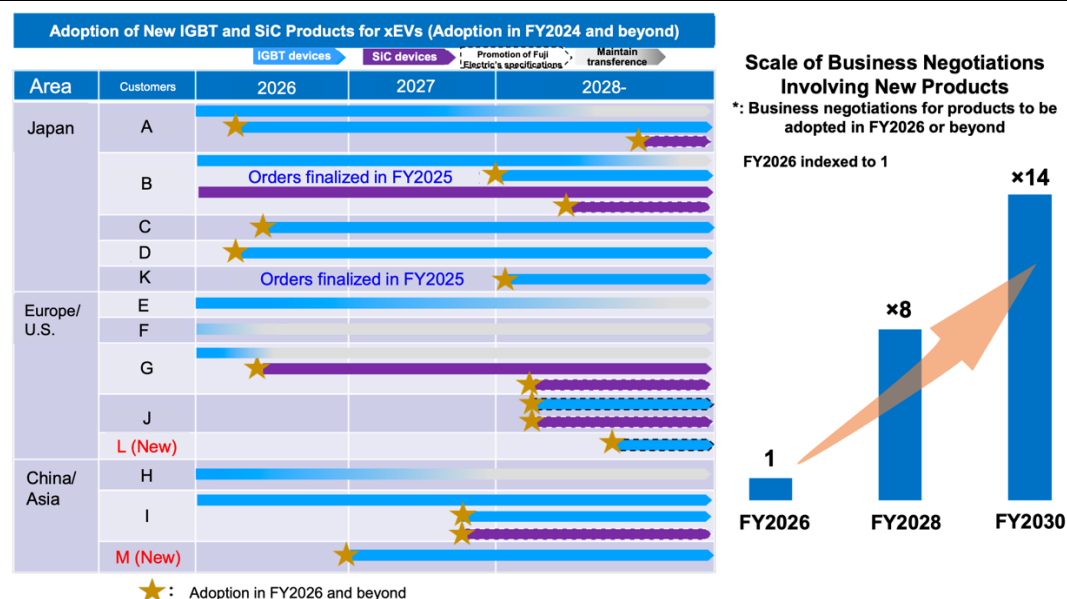
Power semi: The company's 8-inch SiC wafer fab in Surabaya has been completed and is already securing orders, though meaningful volume ramp is anticipated over the medium term. Mitsubishi Electric remains constructive on SiC's long-term prospects, particularly within data center applications. Leveraging its strengths in power module assembly technologies, the company aims to sustain its current global No. 2 market share position, trailing only Infineon. The firm has no immediate plans for price increases and will continue to drive cost reductions to safeguard profit margins.

Partnership developments: 1) Mitsubishi Electric has established a 50/50 joint venture with Hon Hai which will capitalize on Hon Hai's capabilities and networks to supply high-quality electric vehicle platforms from Japan, encompassing powertrain and autonomous driving technologies. 2) In a tripartite collaboration with Toshiba and Rohm, the companies aim to reinforce competitive advantages in the power device business, targeting a scale and technology base that can effectively compete in global markets. This joint venture is expected to generate ¥400 billion in revenue by 2030.

Fuji Electric

Power semi: 1) Auto (accounted for ~50% of FY 2025 revenue): The company's primary auto client base is in Europe and the U.S., with Asia representing only 10% of sales, reflecting a deliberate strategy to avoid competing on price. SiC penetration within its automotive portfolio stood at 10% in FY2025, with guidance rising to 25% in FY2026 and 60% by FY2030. The company benefits from roughly three years of order visibility in automotive, underpinning high revenue predictability. 2) Industrial (~50% of FY2025 revenue): SiC adoption in data center UPS systems currently remains in the single-digit percentage range, though a substantial uptick is anticipated.

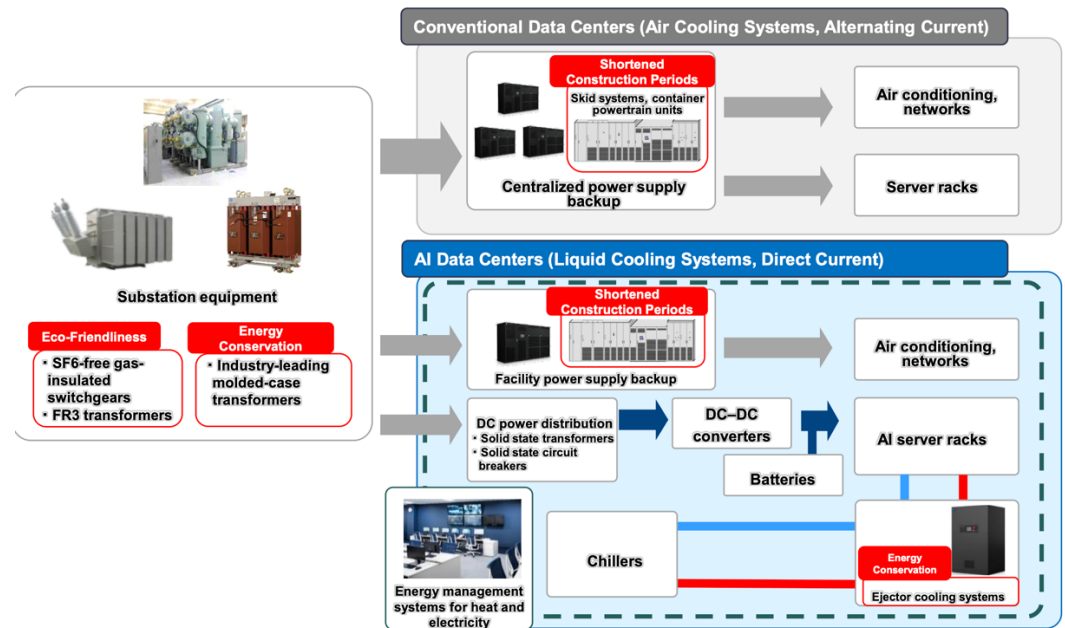
Figure 1: Fuji Electric's auto semi development



Sources: Company Data, GF Securities (Hong Kong) Brokerage.

Datacenter exposure: Fuji Electric projects a revenue CAGR of 32% for its data center-related business over the coming years, materially outperforming the estimated global data center construction CAGR of 18.8%. The company is developing a comprehensive total solution for AI-oriented data centers (including facility power supply backup, air cooling, DC distribution, DC-DC converters, and energy management systems) with MP expected from FY2028.

Figure 1: Fuji Electric's datacenter related products



Sources: Company Data, GF Securities (Hong Kong) Brokerage.

Raw material prices: Rising prices for steel, copper, gold, silver, and other base metals are prompting Fuji Electric to pursue cost pass-through measures. Pricing strategies vary by product segment: 1) For low-voltage industrial inverters, price increases are challenging due to competitive pressure from Chinese manufacturers, while other industrial products have already been subject to price hike notifications in 1H26, with actual implementation scheduled for 2H. 2) For specification-driven products such as automotive, energy, and UPS systems, where Fuji Electric holds stronger competitive positioning, price increases are more readily executable. The company targets restoring auto power device operating profit margins to 15%, and expects a demand recovery in the auto market by 2028 to accelerate margin normalization.

Partnership developments: 1) Fuji Electric is collaborating with Denso on auto IGBTs, with their jointly developed RC-IGBT receiving its first automotive adoption, serving customers including Toyota. 2) The two companies are also co-developing SiC products focused on automotive demand, and are actively expanding 8-inch SiC production capacity, with volume shipments expected to commence in 2028.

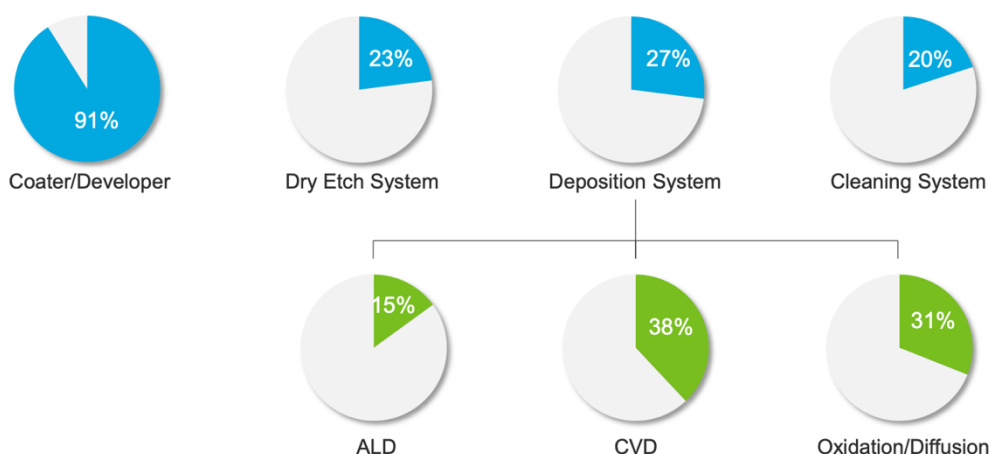
Tokyo Electron (TEL)

Production: Tokyo Electron has increased etching equipment production from one shift to two shifts (with maximum capacity at three shifts), while other production lines continue to operate on a single-shift basis (weekdays only). Lead times for most equipment currently stand at ~5 months, with certain models extending to 6 months. For Chinese customers, lead times are typically 8–10 months, longer than for clients in other regions, as the company typically begins development preparations one generation in advance for large customers in other regions with well-defined specifications.

Price adjustments: TEL increases price in two phases: 1) surcharges on existing models, which certain clients have already accepted and will be reflected in earnings from 2H26; 2) proactive price increases on new models. The company's GPM guidance for 2026 is 45.5%, with a medium-term target of achieving 50% GPM within one to two years. To date, Tokyo Electron has observed no significant upstream raw material or component cost inflation.

Market share dynamics: Etching equipment market share declined by 4ppts last year, primarily attributable to: 1) customer mix effects, particularly Intel's capex reduction; 2) elevated procurement from U.S. peers by Chinese customers; and 3) yen depreciation. The company is confident in regaining share in 2026, supported by its dominant position in DRAM and demand from China.

Figure 1: TEL's world market share in CY2025



Source: Gartner®, Market Share: Semiconductor Wafer Fab Equipment, Worldwide, 2025, Bob Johnson and Menglin Cao, 2 April 2026, Revenue from Shipments basis. Chart created by TEL based on Gartner research. Gartner research. Calculations performed by TEL.
Coater/Developer: Photoresist Processing (Track), Dry Etch System: Dry Etch, Deposition System: Tube CVD + Atomic Layer Deposition Tools + Oxidation/Diffusion Furnaces + Nontube LPCVD, ALD: Atomic Layer Deposition Tools, CVD: Tube CVD + Nontube LPCVD, Oxidation/Diffusion: Oxidation/diffusion Furnaces, Cleaning System: Single Wafer Processors + Wet Stations + Batch Spray Processors + Scrubbers + Other Clean Equipment, Wafer Bonder: Wafer Bonder.
GARTNER is a trademark of Gartner, Inc. and its affiliates. Gartner does not endorse any company, vendor, product or service depicted in its publications, and does not advise technology users to select only those vendors with the highest ratings or other designation. Gartner publications consist of the opinions of Gartner's business and technology insights organization and should not be construed as statements of fact. Gartner disclaims all warranties, expressed or implied, with respect to this publication, including any warranties of merchantability or fitness for a particular purpose. The Gartner content described herein (the "Gartner Content") represents research opinion or viewpoints published, as part of a syndicated subscription service, by Gartner, Inc. ("Gartner"), and is not a representation of fact. Gartner Content speaks as of its original publication date (and not as of the date of this Presentation), and the opinions expressed in the Gartner Content are subject to change without notice.

Sources: Company Data, GF Securities (Hong Kong) Brokerage.

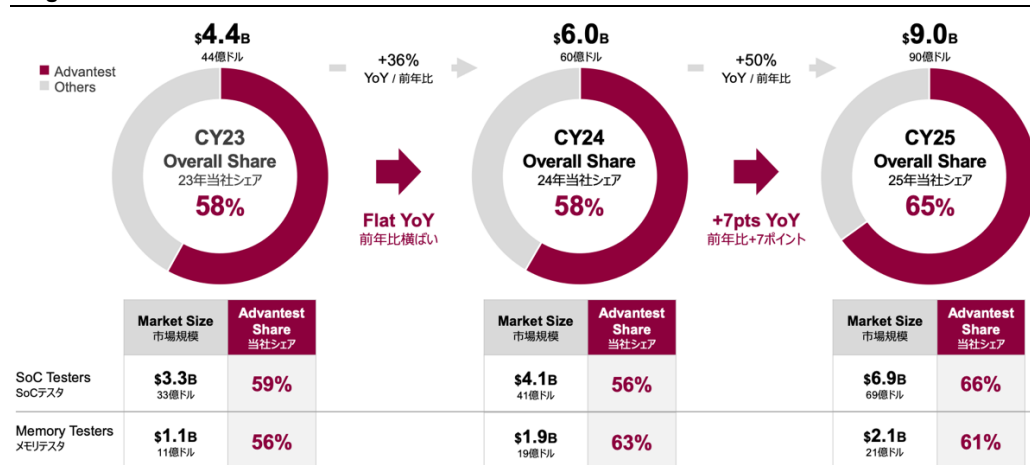
China contribution: China is expected to contribute in the low-30% range of total revenue this year. For China demand, TEL forecasts 20% YoY growth in China WFE in 2026, driven by increased spending from memory (DRAM and 3D NAND), large logic foundries, and emerging customers. The company experienced a revenue decline in China in CY2025, primarily attributable to share loss against Teradyne, TEL expressed strong confidence in regaining share in 2026.

Advantest

AI upcycle with SoC tester capacity doubling in 2H28: Mgmt. stated that AI semiconductors require longer test time, higher test coverage and richer tester configurations, supporting both volume growth and higher content to drive record-high revenue and profit in FY25. Advantest is expanding V93000 SoC tester capacity with EMS partners from 5k annualized units in FY26 to 10k target in 2H28.

Market share and TAM: The company estimates total tester market share rose from 58% to 65% in CY24-25, with SoC tester share rising from 56% to 66%. Advantest estimates SoC tester TAM to be ¥4.1/6.9/8.7-9.5bn and memory tester TAM to be ¥1.9/2.1/2.2-2.7bn in CY24-26, with guidance range from 3-6 months lead time in 2026, limited 2H26 visibility, and capacity ramp.

Figure 1: Advantest's TAM and market share



Sources: Company Data, GF Securities (Hong Kong) Brokerage.

China contribution: The company indicated that testers remain unaffected by export restrictions, and Chinese vendors focus primarily on low-end market while Advantest concentrates in high-end market, seeing limited competition from local vendors.

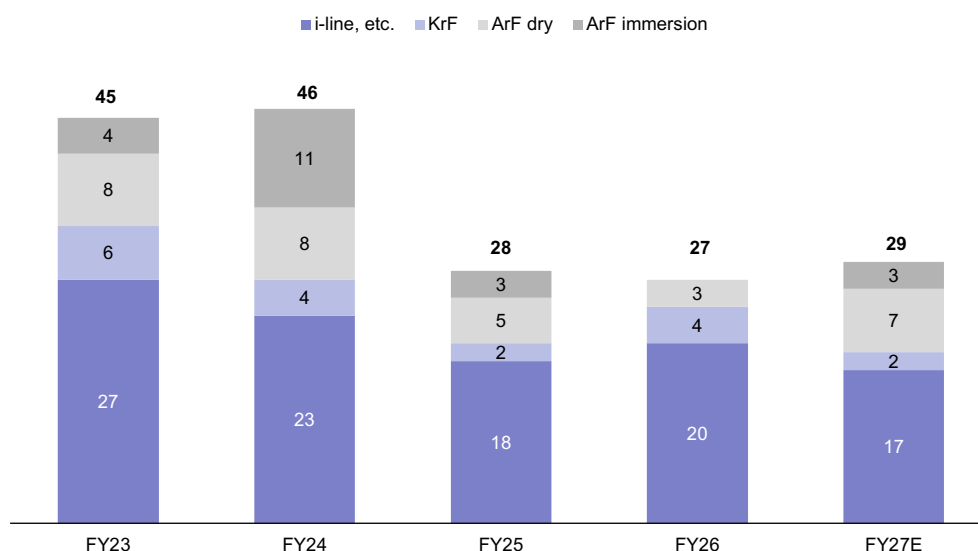
Overall, Advantest guides bullish, while we see margin trends, pricing hikes, and supply chain bottleneck as key watch. Mgmt. indicated that passing through higher costs via price increases remains challenging. Despite constructive revenue outlook and rising SoC mix, which carries higher margin than memory testers, the company guides FY26 GPM at ~63% vs. 64.3% in FY25, due to higher memory costs. That said, the tightness of memory could be the bottleneck when Advantest expands capacity, and others materials such as passives and copper will follow.

Nikon

Lithography growth after 2030: Nikon's mgmt. expects Imaging business to remain the primary profit driver through 2030, with its cash flow investing into Precision Equipment. The company plans to allocate most of ¥800bn R&D and CAPEX budget to next-generation ArF lithography and maskless digital lithography in FY26-30, expecting lithography to become its primary growth driver after 2030, and continues to expand its existing ArF system ahead of new platform launch before that. Meanwhile, Nikon plans to ship its first digital lithography system in 1H27 targeting PLP, with little revenue contribution before 2030, as they believe conventional steppers will remain the mainstream for 310mm wafer-level packaging, while digital lithography to become attractive as panel sizes expand to 500–600mm, where maskless exposure offers better productivity and stitching accuracy.

China contribution: Nikon indicated that ArF dry systems remain unaffected by export restrictions, while advanced ArF immersion systems are not allowed, though downgraded versions are exportable, and expects ArF shipments in China to account for 20-25% over long term. Despite that mgmt. expect the potential opportunities to penetrate more Tier-1 customers as they might seek for second supplier, we remain cautious as Nikon reported no ArF immersion system sales in FY26 and projects only 3 units in FY27, and main customer, Intel, has very little demand for ArF immersion which carries 10x ASP vs. ArF dry and could be the key determinant of lithography business.

Figure 1: Nikon's lithography systems sales by technology



Sources: Company Data, GF Securities (Hong Kong) Brokerage.

Tokyo Ohka Kogyo (TOK)

Market share: TOK estimates market share of photoresist across i-line, KrF, ArF, and EUV to be 22.7%/32.4%/15.9%/28%, ranking 1st/1st/4th/2nd, respectively. Mgmt. noted that share shifts in mature photoresists up to ArF are limited, while **EUV remains the key share-gain opportunity** as 2nm and future advanced nodes ramp, where TOK's materials are adopted by major EUV customers in Taiwan, Korea and the US.

Pricing hike since 3Q26: The biggest cost pressure comes from naphtha solvents used in photoresist formulations due to Middle East tensions. Mgmt. noted price hikes are ongoing, with negotiations starting in May after supplier price-increase notices in April, which precede customer pass-through, creating a timing lag and likely 2Q26 GPM pressure. The pass-through has proceeded relatively smoothly, supported by high customer utilization rates and the reluctance of clients to switch photoresist suppliers.

Capacity expansion: TOK is expanding high-purity chemical capacity, with new production line in 1H27 and new chemical plant in Korea in 2H27. Looking ahead, mgmt. noted that the photoresist consumption remains uncertain, as usage is affected by wafer size and the number of lithography steps at the same time.

China contribution: TOK indicated no export restrictions on its China shipments, and some China customers raised inventory from 2–3 months to 4–5 months after restriction news, leading to a temporary revenue decline in China in 2Q, though the company does not anticipate a material lasting impact on future China sales.

Risks

1) AI demand deceleration; 2) Geo-political uncertainties; 3) Competition.

Rating definitions Benchmark: Hang Seng Index (Hong Kong)

Company ratings	Buy	Stock expected to outperform benchmark by more than 10%
	Hold	Expected stock relative performance ranges between -10% and 10%
	Underperform	Stock expected to underperform benchmark by more than 10%

Sector ratings	Positive	Sector expected to outperform benchmark by more than 10%
	Neutral	Expected sector relative performance ranges between -10% and 10%
	Cautious	Sector expected to underperform benchmark by more than 10%

Hong Kong Company	GF Securities (Hong Kong) Brokerage Limited
Address	27/F, GF Tower, 81 Lockhart Road, Wan Chai, Hong Kong
Telephone	(852) 37191111
Email	michellejing@gfgroup.com.hk

Disclaimer

This report has been prepared by GF Securities (Hong Kong) Brokerage Limited ("GF Securities (Hong Kong) Brokerage"). According to the laws, regulations and regulatory requirements in different countries and regions, this report is distributed by GF Securities (Hong Kong) Brokerage with relevant legal and compliant operation qualifications in these countries and regions.

GF Securities (Hong Kong) Brokerage is licensed by the Securities and Futures Commission of Hong Kong ("SFC") to conduct Type 4 Regulated Activity "Advising on Securities". It is regulated by the SFC, and is responsible for the distribution of this report in Hong Kong. Information about the qualifications of the research analyst(s) who is(are) the author(s) of this report as licensed by the SFC are disclosed in the section where research analyst names are shown.

The research analyst(s) primarily responsible for the content of this report, in whole or in part, certifies that with respect to the company or relevant securities that the analyst(s) covered in this report: 1) all of the views expressed accurately reflect his or her personal views on the company or relevant securities mentioned herein; and 2) no part of his or her remuneration was, is, or will be, directly or indirectly, in connection with his or her specific recommendations or views expressed in this report.

This report is published solely for information purpose and does not constitute an offer to buy or sell any securities or a solicitation of an offer to buy, or recommendation for investment in, any securities.

The securities mentioned in this report may not be allowed to be sold in certain jurisdictions. No action has been taken to permit the distribution of the research report to any person in any jurisdiction that the circulation or distribution of such research report is unlawful. This report is distributed solely to clients or designated institutions authorized by GF Securities (Hong Kong) Brokerage, and is not distributed publicly. It is distributed to certain clients based on the conclusion that they are able to assess investment risks independently, execute investment decisions independently and assume corresponding risks independently.

This report has been issued and based on information obtained from sources generally available to the public and believed by the research analyst(s) to be reliable but which has not been independently verified. No representation or warranty, either express or implied, is made by GF Securities (Hong Kong) Brokerage as to their accuracy and completeness of the information contained in this report. GF Securities (Hong Kong) Brokerage accepts no liability for all loss arising from the use of the materials presented in this report, unless is excluded by applicable laws or regulations. Please be aware of the fact that investments involve risks and the price of securities may be fluctuated and therefore return may be varied, past results do not guarantee future performance. Any recommendation contained in this report does not have regard to the specific investment objectives, financial situation and the particular needs of any individuals. This report is not to be taken in substitution for the exercise of judgment by respective recipients of this report, where necessary, recipients should obtain professional advice before making investment decisions.

GF Securities (Hong Kong) Brokerage may have issued, and may in the future issue, other communications that are inconsistent with, and reach different conclusions from, the information presented in the research report. The points of view, opinions and analytical forecasts adopted in the research report are solely expressed by the analysts but not that of GF Securities (Hong Kong) Brokerage or its affiliates. The information, opinions and forecasts presented in the research report are the current opinions of the analysts as of the date appearing on this material only which may subject to change at any time without notice. The salesperson, dealer or other professionals of GF Securities (Hong Kong) Brokerage may deliver opposite points of view to their clients and the proprietary trading division with respect to market commentary or dealing strategy either in writing or verbally. The proprietary trading division of GF Securities (Hong Kong) Brokerage may have different investment decision which may be contrary to the opinions expressed in the research report. GF Securities (Hong Kong) Brokerage or its affiliates or respective directors, officers, analysts and employees related to research report business may have rights and interests in securities mentioned in the research report. Recipients should be aware of relevant disclosure of interest (if any) when reading this report.

GF Securities (Hong Kong) Brokerage and its affiliates may be seeking or building business relationships with company(ies) mentioned in this report. Therefore, investors should consider the impact on the independence of this report by GF Securities (Hong Kong) Brokerage and its affiliates due to potential conflicts of interests. Investors should not make any investment decisions based solely on the contents of this report. Investors should make their own investment decisions and bear their own risk. No written or verbal commitment of sharing gains or losses from securities investments in any form shall be effective.

This report may contain and/or describe/present factual historical information on prices of Futures contracts (the "information"). Please note that this information is solely for the purpose of forming part of the argument/grounds/evidence in our research methodology/analysis to support our conclusion on our view of the relevant industry/company mentioned. It does not, by any means (express or implied) to be associated with or constituted as SFC Type 5 Regulated Activity (Advising on futures contracts).

Disclosure of Interests

- 1) The research analyst(s) and his/her associate has not served as an officer of the company(ies) mentioned in this report, and does not have any financial interests in the company(ies) mentioned in this report.
- 2) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any financial interests and has not invested in interests aggregate to an amount equal to or more than (i) 1% of the market capitalization; or (ii) 1% of the issued share capital, or issued units, in the company(ies) mentioned in this report.
- 3) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any market making activities, and has not employed any individual(s) serving as officer(s) of the company(ies) mentioned in this report.
- 4) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any investment banking relationship in Hong Kong with the company(ies) mentioned in this report in the past 12 months.

Copyright © GF Securities (Hong Kong) Brokerage

Without the prior written consent obtained from GF Securities (Hong Kong) Brokerage, any part of the materials contained herein should not (i) in any forms be copied or reproduced or (ii) be re-disseminated.